

INDEX

Chapter No	Name OF Topic	Page No
	Certificate	
	Acknowledgement	
	Abstract	
	Index	
	List Of Figures	
1	INTRODUCTION	1
	1.1 Introduction	
	1.2 Need of Project	
	1.3 Objective of Project	
2	LITERATURE SURVEY	4
3	METHODOLOGY	6
4	PERFOMANCE ANALYSIS	11
5	CONCLUSION	13
6	REFERENCES	1

LIST OF FIGURES

Figure No	Name Of figure	Page NO
Fig.1	Block Diagram	6
Fig.2	Solar Panel	7
Fig.3	Battery	7
Fig.4	Arduino	8
Fig.5	Servo Motor	8
Fig.6	LDR	9
Fig.7	Jumper wire	10
Fig.8	Resistor	10

CHAPTER 1

INTRODUCTION

1.1 Introduction:

The shift from conventional fossil-based resources several solar power resources are available in like geothermal energy, wind energy, bio-energy as well as solar power. Solar energy is among the most significant attempting to solve the issue of worldwide heating in addition to is a source of fresh energy. it may also assist in provided that an alternative so sensational fossil-based energy resources in addition to as a cost-effective source of energy besides. Consumption is actually timed enclosed by photovoltaic and thermal energy, making use of solar panels as well as solar collectors, respectively. there will be numerous applications given that photovoltaic thermal sporting goods. At low-temperature ranges, space conditioning and domestic hot water by-product tend to be the overall most typical use. at medium temperature ranges, solar temperature reduction, desalinisation furthermore postindustrial process high temperature tend to purpose which can deed alternative energy. at high-temperature stages, concentrating solar power plants are going to be sensational programs that catch the overall most attention of worldwide.[1]

For any nation to be developed and to progress, energy plays a huge factor in it. The rapid progress in industrialization and urbanization has led to a drastic unbalanced channelization of energy globally. A huge part of energy is withdrawn, administered and enthralled in the global society daily. Fossil fuels are responsible to produce nearly 85 percent of the energy. As we know, fossil fuels are not available in huge amounts, their resources are limited and most importantly, their depletion rate has gone an all-time high, the increasing amount of their usage and consumption is resulting in the rise of the Earth's temperature, which in turn is resulting in a global issue. Journal of Mines, Metals and Fuels, 71(10): 1725-1736; 2023. DOI: 10.18311/jmmf/2023/35863 Global warming. Scientists, researchers, developers, and entrepreneurs have now started to realise the importance of renewable energy sources. Extensive research and study are going on, on how to make the best use of freely available energy sources. Therefore, the demand for solar energy as fuel, geothermal energy, ocean tidal and wave energy has increased[2].

Renewable energy resources provide about 28% of electricity generation across the world. The demand for power generation will keep increasing in the coming years. However, resources like coal, oil, and gases, that is the non-renewable ones are exhausting. The solar energy is one of the most powerful resources for the generation of power. In the last decades, most of the residential areas installed solar systems. Thus, to utilize the efficiency of solar energy, the idea solar tracking system

is proposed. The solar energy produced, largely depends upon the placement and orientation of the panel. To enhance the output obtained, the panel must be perpendicular to the sun throughout the day. As in the existing system. Sun tracking solar panel is a project which aims at designing a model in which the solar panel is self-orienting with respect to the positions of the sun in the sky. The innovation in this phenomenon is that, it generates more amount of power from solar energy than fixed systems in a lesser area. In fixed systems, to produce more power the panels of larger areas are required to be installed, but in this case, more amount of energy is generated while keeping the area of the panel the same. Thus, this method is most cost-effective as well as contributes to the proper utilization of energy.[3]

1.1Need of Project :

A solar panel sun position tracking project is designed to optimize the capture of solar energy by ensuring that the panel remains aligned with the sun's position throughout the day. Unlike fixed solar panels, which remain stationary and only receive maximum sunlight for a limited time, a tracking system adjusts the panel's angle continuously to face the sun directly. This approach significantly increases the efficiency of solar energy collection, making it ideal for educational projects, small-scale renewable energy solutions, and energy-efficient technology demonstrations.

The core of the system is the solar panel itself, which converts sunlight into electrical energy. To ensure maximum output, the panel is mounted on a motorized platform that can tilt and rotate in response to the sun's movement. This movement is typically controlled by a microcontroller (like an Arduino or Raspberry Pi) that receives input from light sensors, such as Light Dependent Resistors (LDRs). These sensors detect the direction of maximum sunlight, and the controller activates the motors to adjust the panel's orientation accordingly.

There are two main types of solar trackers used in such projects: single-axis and dual-axis. A single-axis tracker follows the sun's movement from east to west, while a dual-axis tracker can also adjust for the sun's elevation angle, offering even greater accuracy and energy yield. The choice between these systems depends on the project's complexity, budget, and the desired level of precision. Dual-axis systems are more efficient but require more sophisticated control mechanisms and hardware.

This kind of project not only demonstrates practical applications of renewable energy but also integrates knowledge of electronics, mechanics, and programming. It's a valuable learning experience for students and hobbyists alike, as it encourages hands-on experimentation and problem-solving. Additionally, such projects contribute to sustainability awareness and can be expanded with features like data logging, wireless monitoring, or integration with home energy systems.

For a sun position tracking project, a solar panel plays a crucial role as both a functional component and a power source. The project typically involves adjusting the orientation of the solar panel throughout the day to follow the sun's movement across the sky, thereby maximizing the panel's exposure to sunlight and increasing its efficiency in energy generation. This type of system, often called a solar tracker, can significantly enhance energy output compared to stationary panels. The solar panel used should be lightweight, efficient, and compatible with tracking mechanisms such as servo motors or stepper motors. In many designs, sensors like light-dependent resistors (LDRs) are placed on the panel to detect sunlight intensity from different directions, feeding data to a microcontroller that adjusts the panel's angle accordingly. This dynamic setup not only demonstrates renewable energy principles but also highlights the integration of electronics, programming, and mechanical systems in achieving real-time solar tracking.

1.2 Objective of Project:

The main objective of a solar panel with a sun position tracking system is to maximize the efficiency of solar energy collection by continuously aligning the solar panel with the sun's position throughout the day. Unlike fixed panels, this system adjusts the panel's angle to follow the sun's path, increasing energy output and optimizing performance. It aims to enhance renewable energy utilization, reduce dependence on fossil fuels, and contribute to sustainable and eco-friendly power generation.

1. Increase Power Generation: Improve solar energy capture by keeping the panel perpendicular to sunlight for longer durations.
2. Optimize Efficiency: Reduce energy loss compared to fixed panels.
3. Demonstrate Automation: Showcase the use of sensors or algorithms (like LDRs, microcontrollers, or astronomical tracking) for automatic sun tracking.
4. Sustainability: Promote the use of renewable energy in a more effective way

CHAPTER 2

LITERATURE SURVEY

2. Literature Survey

[1] In the past, research was made to solve the issue of loss of energy by solar panels. This paper suggests that the efficiency of solar power systems can be increased by incorporating a tracking system. Various types of tracking systems are possible. This paper is an overview of designing a single-axis sun tracker using a microcontroller. The study evinces that such tracking systems increase the yield of energy by a huge amount. Using microcontrollers manipulates the execution of the system with more efficiency.

[2] This paper addresses the factors that affect the efficiency of solar cells such as temperature, maximum power point tracking, and energy conversion efficiency. Solar cells are more efficient in various applications if all these factors are optimized. Thus, to utilize these factors a tracking system based on the function of a DC motor controlled by light sensors is fabricated.

[3] The tracking system is designed using a microcontroller as well as a servo motor. The system can be programmed to rotate at different angles and also the sensitivity of the system can be encouraged by the use of Light Depending Resistors that is LDR sensors. This gives greater flexibility over existing systems. The paper explores how the single-axis algorithm could be extended to the dual-axis as well.

[4] This paper proposes a financial assessment of the economic benefits of tracking systems over traditional systems. The financial survey was undertaken based on Texas's energy market rate and sun trackers' average price and operative expenses. The overall improvement in the final outcome was about 82%. Thus, the trackers proved to be economically more beneficial.

[5] This paper aims to the comparison between tracking panels and fixed panel PV systems with the same rate powers in one year. The setup was installed in Florida. The Factors such as the annual power production by each array and the amount of CO₂ emission reduction suggested that the tracker system should be adopted and more research should be carried out in this aspect.

[6] The paper relates to the architecture of a simple dual-axis tracking system using Arduino and a stepper motor. It gives the algorithm of the relation between the working of a motor based on the signals received from LDR which are responsible for the moment of the solar panel.

[7] This paper approaches a photovoltaic panel system using an Arduino-based application as monitoring media. The Android application monitors the temperature changes and conveys the necessary angle inputs. The node MCU is the communication bridge for the microcontroller. The system is commanded by open loop control system. Servo motors are attached to bike wheels which move solar panels to follow the sun's positions. Thus, the output from the mobile application which serves as input for the servo motor and the position of the panel is adjusted.

[8] The tracking systems designed using microcontrollers are efficient as they provide systematic working of the system. The LDR sensors used with appropriate arrangement provide the signals to the servo motor and thus the panel rotates with the desired angle according to the position perpendicular to the sun rays.

[9] The paper performs a techno-economic-environmental analysis of maximum power generation by the solar tracking system. The panels were installed in the residential areas. The results reflected that, by using tracking systems the number of plants needed was reduced. The single-axis tracking system resulted in greater cost-efficiency with 0.243 \$/kWh cost of energy with an average of 25% increase in the output power. Although the dual-axis tracker increased power generation by 33%, it was less cost-effective.

[10] The paper directs to the design of a dual-axis solar tracker. The panel adjusts its orientation by moving simultaneously in two directions. It is a closed-loop system using the Wheatstone bridge circuit that works with light-dependent resistors (LDRs).

CHAPTER 3

METHODOLOGY

3. Methodology:

The methodology for a solar panel sun position tracking system involves designing a setup that allows the panel to follow the sun throughout the day for maximum energy absorption. The project begins with identifying the need for improved efficiency over fixed panels. A microcontroller (like Arduino) is used to process data from sensors (LDRs) or calculate sun angles using algorithms. This data controls motors that adjust the panel's orientation. Components like solar panels, sensors, motors, and a rotatable frame are assembled and programmed. The system is then tested, calibrated, and its energy output compared with a fixed panel setup to evaluate performance gains.

3.1 Block Diagram:

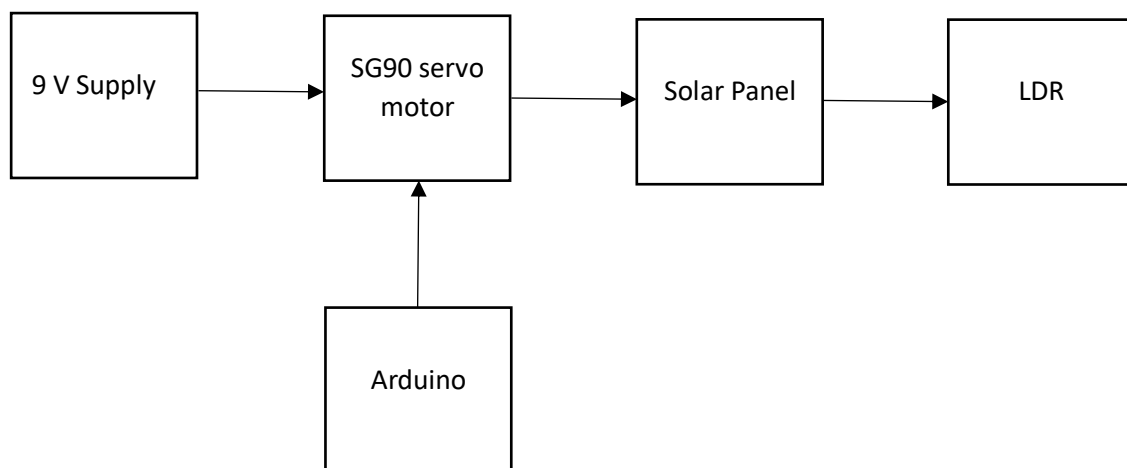


Fig.1. Block Diagram

3.2 Component Selection:

Component selection is vital for the efficient and reliable operation of the home automation system. The chosen components ensure seamless integration, power efficiency, and safe switching operations.

1.Solar Pannel:

A solar panel project involves selecting suitable solar modules based on energy requirements, site analysis for optimal placement, and designing an efficient system layout. After calculating load and energy needs, components like solar panels, batteries, charge controllers, and inverters are chosen. The panels are installed at an angle to maximize sunlight exposure. Electrical connections are made following safety standards, and the system is tested for voltage, current, and energy output.



Fig 2 : Solar Panel

2.HW battery:

A hardware battery used to identifying the energy storage requirements and selecting the appropriate battery type (e.g., Li-ion, lead-acid, or NiMH) based on capacity, voltage, and application.. Hardware components like voltage sensors, protection circuits, and charge controllers are connected. The battery is then charged, tested under different loads, and monitored for performance metrics such as voltage stability, discharge rate, and temperature.



Fig 3: Battery

3.Arduino:

An Arduino-based project starts with defining the problem and determining the system requirements. Suitable sensors, actuators, and modules are selected based on the application. The Arduino board is programmed using the Arduino IDE, with code written to read inputs, process data, and control outputs. Components are connected to the board through digital and analog pins following a circuit diagram. The system is assembled on a breadboard or PCB, tested for functionality, and debugged to ensure correct operation. Finally, performance is evaluated and optimized for stability, accuracy, and efficiency.

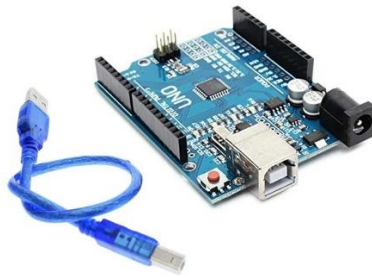


Fig 4 : Arduino

4.SG90 servo motor:

An SG90 servo motor begins with understanding its control mechanism, which relies on PWM (Pulse Width Modulation) signals. The servo is connected to a microcontroller like Arduino, with the control wire attached to a PWM-capable digital pin and powered using 5V and GND. The system is programmed to send specific pulse widths (typically 1–2 ms) to set the desired angle (0° to 180°). Calibration is done to ensure accurate movement. The motor is then integrated into the mechanical setup, tested for range, speed, and response, and fine-tuned based on application needs like positioning, rotation, or tracking.



Fig 5 : Servo Motor

5. LDR Module:

An LDR (Light Dependent Resistor) module begins with selecting the appropriate LDR sensor based on the light sensitivity range needed for the application. The LDR is typically connected in a voltage divider circuit, where one end is connected to a known resistor and the other end is connected to the analog input pin of a microcontroller (such as Arduino). The LDR's resistance changes in response to the intensity of light, which can be measured by the microcontroller as a varying voltage signal. The system is programmed to read this signal, convert it to a usable value, and trigger actions based on light levels (e.g., turning on lights, controlling a solar panel's position, or adjusting screen brightness). Calibration is done to adjust the threshold values for specific light conditions. The circuit is tested under different lighting conditions to ensure the system responds accurately to changes in light intensity.



Fig 6 : LDR

6.Jumper wire:

Jumper wires in electronics projects involves selecting the appropriate jumper wire length and type (male-to-male, male-to-female, or female-to-female) based on the connections needed. Jumper wires are used to establish temporary, flexible electrical connections between components such as sensors, microcontrollers (like Arduino), power sources, and other devices on a breadboard or PCB. The process starts by identifying the pins or components that need to be connected. Jumper wires are then inserted into the corresponding pins of the components or the breadboard, ensuring proper alignment of signal, ground, and power connections. The connections are checked for proper contact and integrity to avoid loose connections or short circuits.



Fig 7 : Jumper Wire

7.Resistor :

Resistors are used in circuits for signal conditioning, adjusting the strength of signals in amplifiers and filters. They also play a key role in feedback networks, controlling the gain in operational amplifiers. Additionally, resistors are essential in timing circuits, where they help control delays or frequencies, particularly in RC circuits.



Fig 8 : Resistor

CHAPTER 4

PERFOMANCE ANALYSIS

4. Performance Analysis:

1. Increased Energy Output

- Solar tracking systems can increase energy generation by 25% to 40% compared to fixed panels.
- Panels are always aligned perpendicular to sunlight, maximizing irradiance.

2. Efficiency Optimization

- Reduces the cosine losses by maintaining optimal angle.
- Ensures higher panel efficiency throughout the day.

3. Real-time Position Adjustments

- Tracks the sun's azimuth and elevation dynamically.
- Uses sensors or programmed algorithms (e.g., astronomical calculations) for movement.

4. Higher Initial Cost

- Tracking mechanisms (servo motors, actuators, sensors, microcontroller) add to the system cost.
- Maintenance and repair can also be higher due to moving parts.

5. Better ROI Over Time

- Despite higher upfront cost, the increased energy production shortens the payback period.
- More suitable in areas with high electricity rates or off-grid applications.

6. Types of Trackers and Their Impact

- Single-axis trackers: Follow the sun east to west; simpler and less costly.
- Dual-axis trackers: Follow both daily and seasonal movements; offer maximum output.

7. Weather Dependency

- Performance gain is higher in sunny climates.
- In cloudy or diffused light conditions, gain is limited compared to clear-sky days.

8. Space Efficiency

- Trackers may require more spacing to avoid shadowing as they tilt throughout the day.
- May reduce panel density in solar farms.

9. Power Consumption of Tracker

- Tracking systems consume a small portion of the generated power (~1-2%).
- Needs to be accounted for in net efficiency calculations.

CHAPTER 5

CONCLUSION

5. Conclusion

It is evident from the above study the use of solar trackers will certainly increase the efficiency of SPV systems but proper care is required while installing solar trackers with these systems. Problems such as failure of solar tracker also need equal attention while installing them. Whether to install active or passive solar trackers is also a question which needs to be explored further. This study also tried to analyze all the important factors which are required for the optimization of solar trackers.

Thus, looking at all sorts of tracking systems, comparing their numerical results and efficiencies with traditional and static solar panels, we can say that the extraction of more power from the solar panel will result in decrease of the cost per watt. This will ensure that extracting power using the tracking system will be much more cost efficient than traditional static solar panels. Thus, we can conclude that, single axis tracking system proves to be more pocket friendly and more flexible in installation as compared with dual axis trackers.

The project helps in enhancing the efficiency of solar panels and thus the maximum amount of energy can be obtained without loss. Similarly, the project shows the working of software for maximizing the output by positioning the panel at maximum intensity. Self-adjusting solar panels are proven useful as we just need to place them in ample sunlight and rest it works efficiently. Hence, the tracking systems demand more research as it proves to be beneficial in agriculture as well as for power generation in industries as well as in residential areas. It is cost-effective as well as environmental-friendly

The project "Manufacturing of Passive Solar Tracking System" aims to develop an innovative and efficient solar tracking system that maximizes the capture of solar energy. The report has provided an overview of the project, starting with the identification of the problem of inefficient solar energy utilization and the need for a passive solar tracking solution. The suggested solution of tracking system without any use of energy that developed from PV panel is a better aggregate which provides 15% more energy generation as predicted in review. Overall, the project offers a promising solution for optimizing solar energy capture through passive solar tracking. The report provides a comprehensive understanding of the project's design, manufacturing, and potential future developments.

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